

Part A: Literature Review and Dataset

Title	Value, Risk, and Recoverability: An Interpretable Order-Level Prioritization Framework for Service Recovery in E-Commerce
Harvard reference	Madane, Y. and Azeroual, M. (2026) 'Value, Risk, and Recoverability: An Interpretable Order-Level Prioritization Framework for Service Recovery in E-Commerce', <i>Digital</i> , 6(3), 57. Available at: https://doi.org/10.3390/digital6030057 (Accessed: 24 September 2026).
Link	https://www.mdpi.com/2673-6470/6/3/57
Modelling	Logistic Regression, Random Forest and Gradient Boosting. The final risk model used calibrated Gradient Boosting, with SHAP used to explain predictions.
Preprocessing techniques	The Olist tables were joined at order level. Missing records were removed, extreme values were trimmed, and features such as delivery time, freight ratio and customer–seller proximity were created.
Datasets used	Brazilian E-Commerce Public Dataset by Olist , covering roughly 100,000 orders from 2016–2018. The study used 93,663 delivered orders for its risk analysis.
Key takeaways	1. Olist can be transformed into a useful order-level predictive dataset . 2. Delivery-related variables can provide useful risk signals. 3. Use a time-based split to make the prediction more realistic.

Title	Case study on delivery time determination using a machine learning approach in small batch production companies
Harvard reference	Rokoss, A., Syberg, M., Tomidei, L., Hülasing, C., Deuse, J. and Schmidt, M. (2024) 'Case study on delivery time determination using a machine learning approach in small batch production companies', <i>Journal of Intelligent Manufacturing</i> , 35, pp. 3937–3958. Available at: https://doi.org/10.1007/s10845-023-02290-2 (Accessed: 25 September 2026).
Link	https://link.springer.com/article/10.1007/s10845-023-02290-2
Modelling	SVM, ANN, XGBoost, Decision Tree, Random Forest and Linear Regression. Grid search, feature reduction and time-series cross-validation were used.
Preprocessing techniques	Duplicate records were removed and missing values were handled using MICE and KNN regression. Dates were converted into business-day values and domain-specific features were created.
Datasets used	Real historical data from two German small-batch manufacturing companies .
Key takeaways	1. Prediction should be made as early as possible using information available at that stage. 2. Feature engineering can improve prediction quality. 3. Use chronological validation to avoid using future information.

Title	<i>Agent-Based Simulation Model for Evaluating Urban Freight Policy to E-Commerce</i>
Harvard reference	Alves, R., da Silva Lima, R., Custódio de Sena, D., Ferreira de Pinho, A. and Holguín-Veras, J. (2019) 'Agent-Based Simulation Model for Evaluating Urban Freight Policy to E-Commerce', <i>Sustainability</i> , 11(15), 4020. Available at: https://doi.org/10.3390/su11154020 (Accessed: 28 September 2026).
Link	https://www.mdpi.com/2071-1050/11/15/4020
Modelling	An agent-based simulation model was developed using customers, carriers, e-commerce stores and delivery lockers as interacting agents. Four delivery scenarios were simulated.
Preprocessing techniques	Input parameters were gathered from e-commerce industry data, previous research and transport providers. Customer locations and delivery characteristics were represented in the simulation model.
Datasets used	E-commerce and logistics information from Ebit, previous studies and local transport providers . The study also used a GIS-based representation of the study area.
Key takeaways	1. Failed first deliveries can lead to significant redelivery activity. 2. Delivery failures have operational and cost impacts. 3. Last-mile factors such as delivery location and customer availability are important when planning deliveries.

Name	Brazilian E-Commerce Public Dataset by Olist
Harvard reference	Olist and Sionek, A. (2018) <i>Brazilian E-Commerce Public Dataset by Olist</i> . Kaggle. Available at: https://doi.org/10.34740/KAGGLE/DSV/195341 (Accessed: 27 September 2026).
Link	https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce
Number of instances	99,441 orders . The dataset contains 9 linked CSV files rather than separate training and testing splits. The main orders table contains 99,441 records.
Number of feature columns	52 original columns across the 9 related datasets . These cover orders, customers, products, sellers, payments, reviews and geolocation.
Number of classes (if applicable)	2 classes for the proposed classification target: Late delivery and Not late . The target will be created from the actual and estimated delivery dates.
Target range (if applicable)	Not applicable , because this is a classification problem rather than regression.
Origin	The data comes from Olist, a Brazilian e-commerce marketplace . It contains real commercial transactions from 2016–2018 , covering orders made through multiple marketplaces in Brazil. The data has been anonymised and includes information about orders, customers, products, sellers, payments, freight and delivery.